Column 3 has no pivots. ∴ By defn, x_3 is a free variable.
By Theorem 2, the system of eqns. has infinitely many solutions.
∴ A nontrivial soln. to the matrix equation $Ax=0$ exists.

~~When h=6, the system of eqns~~

$\boxed{\text{So, when } h=6,\ v_1, v_2 \text{ and } v_3 \text{ are linearly dependent}}$

$\boxed{\text{We can similarly prove that when } h\neq 6,\ v_1, v_2 \text{ and } v_3 \text{ are linearly independent.}}$ □

13. $\begin{bmatrix}1\\5\\-3\end{bmatrix}, \begin{bmatrix}-2\\-9\\6\end{bmatrix}, \begin{bmatrix}3\\h\\-9\end{bmatrix}$ $\boxed{\text{Solution:}}$ By defn, $\exists\, c_1, c_2, c_3 \in \mathbb{R}$ s.t. not all of c_1, c_2 and c_3 are 0 together and $c_1v_1+c_2v_2+c_3v_3=0$

∴ $c_1\begin{bmatrix}1\\5\\-3\end{bmatrix}+c_2\begin{bmatrix}-2\\-9\\6\end{bmatrix}+c_3\begin{bmatrix}3\\h\\9\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$. By defn. of matrix equation $Ax=b$, we have: $\begin{bmatrix}1&-2&3\\5&-9&h\\-3&6&9\end{bmatrix}\begin{bmatrix}c_1\\c_2\\c_3\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$.

By Theorem 3, the matrix equation $Ax=b$ has the same solution set as the system of equations corresponding to the augmented matrix $\begin{bmatrix}A & b\end{bmatrix}$.

$$\begin{bmatrix}1&-2&3&0\\5&-9&h&0\\-3&6&9&0\end{bmatrix}\xrightarrow[-5\times Row1]{Row2=Row2}\begin{bmatrix}1&-2&3&0\\0&1&h-15&0\\-3&6&9&0\end{bmatrix}\xrightarrow[+3\times Row1]{Row3=Row3}\begin{bmatrix}1&-2&3&0\\0&1&h-15&0\\0&0&18&0\end{bmatrix}$$

$$\xrightarrow[\times 1/18]{Row3=Row3}\begin{bmatrix}1&-2&3&0\\0&1&h-15&0\\0&0&1&0\end{bmatrix}\xrightarrow[-3\times Row3]{Row1=Row1}\begin{bmatrix}1&-2&0&0\\0&1&h-15&0\\0&0&1&0\end{bmatrix}\xrightarrow[-(h-15)\times Row3]{Row2=Row2}$$

$$\begin{bmatrix}1&-2&0&0\\0&1&0&0\\0&0&1&0\end{bmatrix}\xrightarrow[+2\times Row2]{Row1=Row1}\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\end{bmatrix}$$

. From Example 2, we know the final matrix is in rref.

Columns of A in rref all have pivots. ∴ By defn, there are no free variables.
∴ By Theorem 2, the system of equations has a unique (trivial) soln.
∴ For all h, $\begin{bmatrix}1\\5\\-3\end{bmatrix}, \begin{bmatrix}-2\\-9\\6\end{bmatrix}, \begin{bmatrix}3\\h\\-9\end{bmatrix}$ are linearly independent. □

∴ No h exists for which the given vectors are linearly dependent. □